

ADITYA CHATURVEDI

Machine Learning Engineer

New Delhi, India | chaturvediaditya910@gmail.com | +91 63972 65125 | linkedin.com/in/adityachaturvedii | github.com/adityachaturvedii

PROFESSIONAL SUMMARY

Machine learning engineer building production AI systems across agentic automation, computer vision, and healthcare. Experience spans model development, multimodal agents, inference services, distributed training, data pipelines, and cloud and on-premises deployment. Work combines applied ML with systems engineering, from research and validation through integration into production software.

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Java, TypeScript

Machine Learning: PyTorch, MONAI, scikit-learn, TensorFlow, Hugging Face, MLX; computer vision, 3D segmentation, multimodal agents, LLMs, radiomics

Systems & Infrastructure: FastAPI, Docker, Redis, React/Next.js, SQLite, Git, AWS (EC2, S3, SageMaker), MLflow, SLURM/HPC, Node.js

Data & Imaging: DICOM/NIfTI, DICOMweb/DIMSE, image registration, PyRadiomics, 3D Slicer, clinical data pipelines

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Consultant

QAI, Antler-backed

June 2025 – March 2026

Germany, Remote

- Build core components of an open-source AI quality-assurance platform that converts natural-language test goals into automated web and mobile execution using multimodal agents.
- Extended DroidRun into a multimodal Android agent combining visual reasoning, accessibility-tree interaction, trajectory capture, artifact upload, and knowledge-graph context during execution.
- Designed UI-flow graph versioning and diffing so test plans are regenerated only when application behavior changes, avoiding unnecessary replanning across repeated runs.
- Built the production feedback loop across a TypeScript telemetry SDK and Python anomaly engine, then added agent-based replay so human alerts are raised only after a suspected failure is independently reproduced.
- Decoupled execution and storage from GCP behind shared abstractions and moved graph persistence into the collaboration service, enabling full local deployment and supporting the platform's open-source release.

Machine Learning Engineer & Project Manager, Consultant

Proxmed Pty Ltd

Oct 2023 – Present

Melbourne, Australia, Remote

- Built an on-premise acute-stroke imaging platform integrating three AI inference services for NCCT and CT perfusion behind a unified FastAPI interface, with Dockerized deployment and a React/TypeScript clinical workflow.
- Designed a resource-aware scheduler with CPU, RAM, disk, GPU, and VRAM admission controls, per-engine concurrency limits, restart recovery, bounded retries, and fail-safe handling of GPU telemetry loss.
- Implemented multi-tenant authorization, RBAC, TOTP MFA, audit logging, PACS/DICOM ingestion, and automatic modality routing from image receipt through inference and result delivery.
- Developed a 3D SegResNet pipeline for early ischaemic hypodensity segmentation on non-contrast CT, achieving mean Dice 0.84 on held-out cases; built preprocessing, registration, post-processing, and distributed HPC training workflows.
- Lead imaging operations for the TENACITY multi-site stroke trial, coordinating sponsor, CRO, imaging-vendor, and internal teams while maintaining a single operational view of screening, data receipt, QC status, and open queries.

Machine Learning Engineer, Contract
Sorcova Health SAS

2025 – Present
France, Remote

- Design and build the Health Profile Service, the core data and inference backend for a precision-medicine platform pursuing EU MDR Class IIa certification.
- Implemented de-identification and governance workflows that preserve model-development utility while supporting GDPR, EU MDR, and EU AI Act requirements.
- Automated generation of legal and quality-system documents, including DPAs, SCCs, NDAs, IP assignments, and audit packs, through a programmatic Node.js document pipeline.

Research Engineer, Clinical AI, Consultant
Institute of Liver & Biliary Sciences (ILBS)

September 2025 – August 2026
New Delhi, India

- Lead a CT radiomics study in alcoholic hepatitis (n=513), using IBSI-compliant feature extraction, pre-registered hypotheses, permutation testing, and multiple-comparison control to evaluate imaging biomarkers.
- Built an LLM-based extraction pipeline using structured outputs to convert free-text hepatology discharge summaries into schema-validated records for cohort analysis.
- Developing a laboratory-free, sex-specific risk model for steatotic liver disease using data from 6,000+ South Asian adults, intended for screening settings without bloodwork.
- Principal AI contributor to “Project Walk For Life,” a MeitY-funded national preventive healthcare initiative leveraging AI, wearable technologies, and digital diagnostics.
- Designing scalable data infrastructure and model-governance frameworks to support deployment-ready clinical AI systems.
- Driving collaborations with academic institutions, startups, and global AI organizations to strengthen ILBS’s translational medical AI research program.

Big Data Intern
MedTourEasy

Dec 2022 – Feb 2023
New Delhi, India

- Built automated ETL pipelines for multi-gigabyte daily healthcare datasets and predictive models in Python and R, replacing manual data-cleaning and validation steps.

SELECTED RESEARCH & PUBLICATIONS

- Co-author, ATLAS individual-patient-data meta-analysis of endovascular thrombectomy in large-core ischaemic stroke, *The Lancet*, 2026.
- Co-author, non-contrast CT hypodensity segmentation study for early ischaemic change, manuscript under review.
- Additional research in portal hypertension, liver biopsy, and CT radiomics, including work submitted or in preparation for clinical conferences and journals.

EDUCATION

B.S., Data Science & Applications
Indian Institute of Technology Madras

2020 – 2024
Chennai, India

B.Tech., Computer Science & Engineering
Lovely Professional University

2020 – 2024
Punjab, India

SELECTED RECOGNITION

- Research Presenter, UNSW Workshop (2025), hypodensity segmentation for early ischaemic change on non-contrast CT.
- First Prize, Deep Learning Hackathon (2023), INR 20,000, for a medical AI solution.